Today's agenda

- Pow(a, n)
- TC & SC of Recursion

AlgoPrep

Q) Given a and N, calculate a^n.

Ex:

a	n	
2	3	→ 8
3	5	→ 243

```
int Pow (int a, int n){
1 if (n==1) { return a; }

2 int temp = Pow (a, n-1);

3 return temp*a;

}
```

Faith: Given a and n, calculate and return a^n.

Main logic:

$a^n \rightarrow temp * a$

$\downarrow$

$a^{n-1} \rightarrow temp$

Base case:

if (n==1) { return a; }

Pow (2,4) 123 ↑ 16

↓ ↑ 8

Pow (2,3) 123 ↑ 4

↓

Pow (2,2) 123 ↑ 2

↓

Pow (2,1) 1

T.C. of 1 function: $O(1) \rightarrow x$

No. of function: N. $\rightarrow y$

T.C of recursion: $x * y = O(1) * N$

$= O(N)$

S.C of recursion: $O(1) * N = O(N)$

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// do something better than O(n).

→ $a^n = a^{n-1} * a$

↳ $a^n = a^{n/2} * a^{n/2}$ if n is even

ex: $2^8 = 2^4 * 2^4$

$2^9 = 2^4 * 2^4 \times\times$

↳ $a^n = a^{n/2} * a^{n/2} * a$ if n is odd

$2^9 = 2^4 * 2^4 * 2$

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```
int Pow (int a, int n){
  if (n==1) { return a; }

  int temp = Pow(a, n/2);
  if (n%2==0){return temp * temp}

  else {return temp * temp * a;}
}
```

Faith: Given a and n, calculate and return a^n.

Main logic:

a^n → n is even = temp * temp

a^n → n is odd = temp * temp * a

$a^{n/2}$ → temp

Base case:

```
if (n==1) { return a; }
```

```
int Pow (int a, int n){
1 if (n==1) { return a; }

2 int temp = Pow(a, n/2);

3 { if (n%2==0){return temp * temp}
  { else {return temp * temp * a;}

}
```

$a = 2$ $N = 37$

262144 * 262144 * 2

Pow (2, 37) 123

↓ 262144

Pow (2, 18) 123

↓ 512

Pow (2, 9) 123

↓ 16

Pow (2, 4) 123

↓ 4

Pow (2, 2) 123

↓ 2

Pow (2, 1) 1

overall T.C:

T.C of 1 function: $O(1)$

No. of function: $O(\log_2 N)$

$\simeq O(\log N)$

overall S.C: $O(1) * \log N \simeq O(\log N)$

if n is negative

$2^{-37} \Rightarrow 2^{37} \rightsquigarrow$ recursive sol^n → ans

⇓

return $\frac{1}{ans}$

→ $a^n = a^{n/2} * a^{n/2}$ → $O(\log N)$ → most optimal here.

$a^n = a^{n/3} * a^{n/3} * a^{n/3}$

$a^n = a^{n/4} * a^{n/4} * a^{n/4} * a^{n/4}$

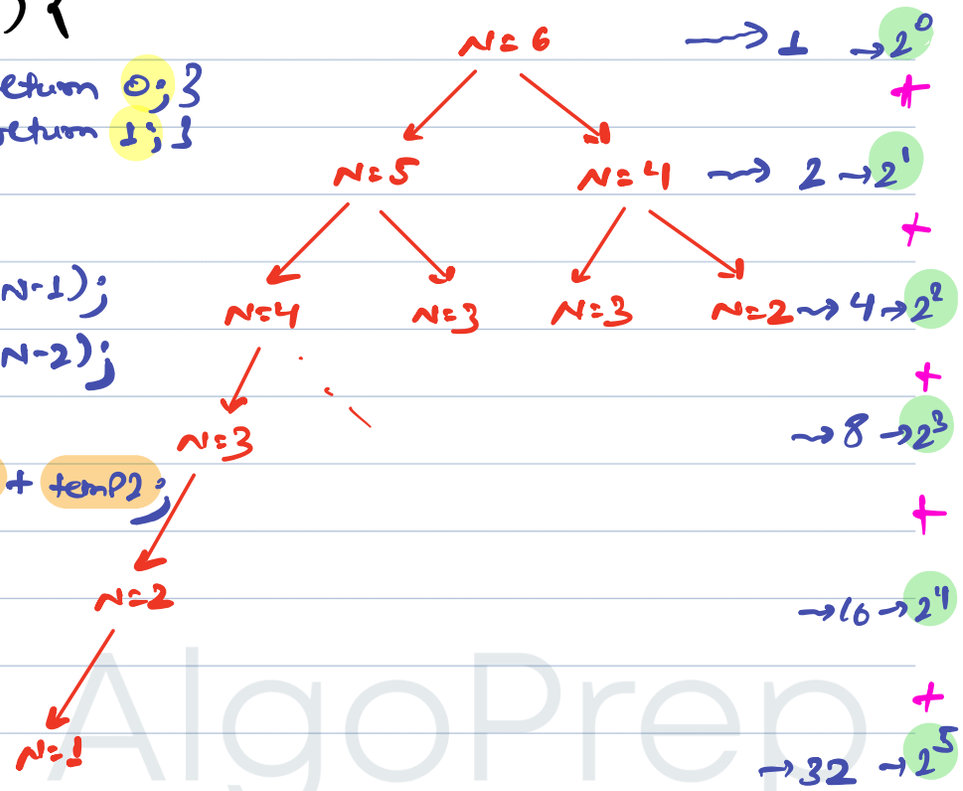
$a^n = a^{n/5} * a^{n/5} * a^{n/5} * a^{n/5} * a^{n/5}$

$a^n = a^{n/n} * a^{n/n} * a^{n/n} * a^{n/n} \ldots\ldots a^{n/2}$ → $O(N)$ → ideal

```
int fib(int N) {
1 { if (N==0){ return 0; }
  { if (N==1) { return 1; }

2 int temp1 = fib(N-1);
3 int temp2 = fib(N-2);

4 return temp1 + temp2;
}
```



$$2^0 + 2^1 + 2^2 + 2^3 \; - - - - - + 2^{N-1}$$

$*2 \quad *2 \quad *2$

$$\text{Sum of G.P} = a * \frac{(r^n - 1)}{r-1}$$

$$= 1 * \frac{2^N - 1}{2-1} = 2^N - 1$$

Overall T.C: $O(1) * 2^N \simeq O(2^N)$

* No. of Calls = x
Count of levels = n no. of functions = x^n

Break till 9:49 pm.

AlgoPrep

Q) Given an array, check if it is Palindrome or not?
↳ recursion

Ex: MALAYALAM → true

Ex: a a b b a → false

//idea 1

→ MALAYALAM
→ MALAYALAM → true

a a b b a → false
a b b a a

//idea 2

0 ↓ ↓ n-1

M A L A Y A L A M

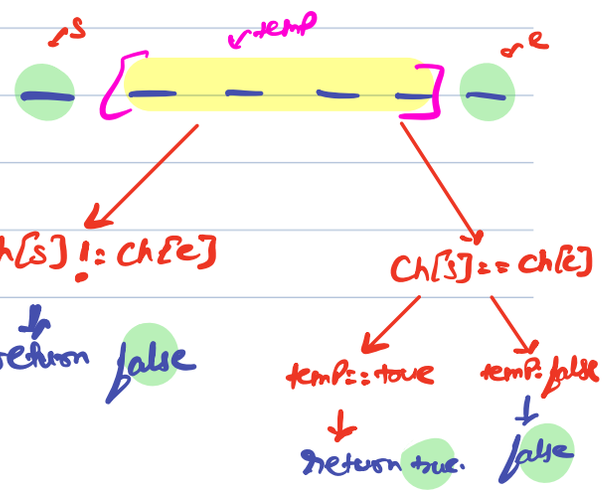
T.C: $O(1) * N/2 \approx O(N)$
S.C: $O(1) * N/2 = O(N)$

```
boolean isPalindrom(char[] ch, int s, int e){
    if (s==e){return true;}
    if (s>e){return true;}

    if (ch[s] == ch[e]){
        boolean temp = isPalindrome(ch, s+1, e-1);
        return temp;
    }
    else{return false;}
}
```

Faith: Check and return whether ch array is Palindrome betn 0 (s) to n-1 (e).

main logic:



base case:

```
boolean isPalindrom(char[] ch, int s, int e){
  1 { if (s == e) {return true;}
    { if (s > e) {return true;}
  if (s>=e) {return true;}

  if (ch[s] == ch[e]){
2   a boolean temp = isPalindrome(ch, s+1, e-1);
    b return temp;
  }
3 else {return false;}
}
```

Ch: M A L A Y A L A M (0 1 2 3 4 5 6 7 8)

isPalindro(0 8) 1 2a b — temp: true — true

↓ true

isPalindrome(1 7) 1 2a b

↓ true

isPalindrom(2 6) 1 2a b

↓ true

isPalindrome(3,5) 1 2a b

↓ t

isPalindrome(4,4) 1

Ch: M A L A A L A M (0 1 2 3 4 5 6 7)

s e

isPalin(0,7) — true

↓ true

isPalind(1,6)

↓ true

isPalind(2,5)

↓ true

isPalin(3,4)

↓ true

isPalin(4,3)

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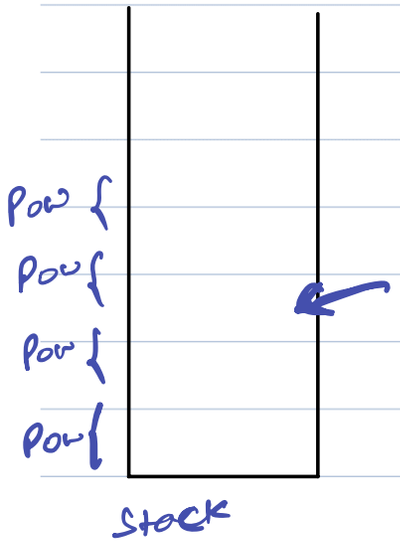
Ch: M A L A A L B M
0 1 2 3 4 5 6 7

false
isPalin (0, 7)
false
isPalin (1, 6)

write iterative code → for (int i=0; i<N; i++) {

}

write recursive code: → within 1 function space complexity is like iteration.
↳ you create multiple instance of same function.



Space will be considered in space complexity.

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S.C: (space used in 1 function) & (Max no. of function in stack at any Point of time.)

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